

Localization of the source of terrestrial neutron bursts detected in thunderstorm atmosphere

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Received 10 August 2009; revised 12 October 2009; accepted 17 November 2009; published 22 May 2010.

[1] To localize an altitude of the neutron source responsible for the neutron flux enhancements observed on the ground, numerical simulations of photonuclear production and transport to on-ground detector locations were carried out. The neutron fluence calculated for the volumetric source located at the altitudes 8–12 km is consistent with that estimated from neutron numbers measured on the ground. This altitude range is consistent with the idea that the burst of hard γ rays detected recently by Tsuchiya et al. (2007, 2009) originate from a volumetric intracloud γ ray source. Most likely, the photonuclear reactions caused by bremsstrahlung of descending relativistic runaway electron avalanches account for the neutron flux increases observed in the thunderstorm atmosphere.

Citation: Babich, L. P., E. I. Bochkov, I. M. Kutsyk, and R. A. Roussel-Dupré (2010), Localization of the source of terrestrial neutron bursts detected in thunderstorm atmosphere, *J. Geophys. Res.*, 115, A00E28, doi:10.1029/2009JA014750.

1. Introduction

[2] In 1925 Wilson pointed out the possibility of nuclear reactions in thunderstorm electric fields due to collisions of accelerated high-energy ions with background thermal nuclei [Wilson, 1925]. A half century later the first experiment was executed under the suggestion that nuclear synthesis reactions $^{12}\text{C}(^2\text{H}, n)^{13}\text{N}$ and $^{14}\text{N}(^2\text{H}, n)^{15}\text{O}$ occur in the lightning channel, however, the results of neutron flux increase measurements in times of thunderstorm activity appeared to be negative [Fleischer, 1975]. It was not until 1985 that a positive result was obtained. In an experiment lasting three years and conducted with a refined configuration in a high mountainous region (the Himalayas, altitude $z = 2743$ m), Shah et al. [1985] were able to measure statistically significant neutron flux increases above background and correlated with lightning electromagnetic pulses (EMP). Later statistically significant increases in neutron fluxes were observed near sea level during thunderstorms [Shyam and Kaushik, 1999; Kuzhevskij, 2004; Gusev et al., 2009] and terrestrial neutron flashes were also detected aboard an artificial satellite of the Earth [Bratolyubova-Tsulukidze et al., 2004].

[3] Following Wilson [1925] and Fleischer [1975], the neutron flux increases reported in those publications were interpreted as the result of nuclear reactions in the lightning channel; namely, due to the synthesis reaction $^2\text{H}(^2\text{H}, n)^3\text{He}$. However, as was recently demonstrated [Babich, 2006, 2007; Babich and Roussel-Dupré, 2007], nuclear synthesis is absolutely impossible under physical conditions met in the lightning channel. Instead, because thunderstorm activity

is accompanied by hard γ emission [Fishman et al., 1994; Smith et al., 2005] with a spectrum stretching above 20 MeV [Smith et al., 2005; Tsuchiya et al., 2007, 2009], (γ, n) reactions are responsible for neutron generation. This result disagrees with the idea of Wilson on collisional nuclear reactions, however, confirms his idea on electron energization up to extremely high energies, actually, above the threshold of photonuclear reactions (γ, n) in the atmosphere. Cosmic rays [Dorman, 2004, 2006] produce in the atmosphere a continuous flux of seed high-energy electrons that in a thunderstorm electric field initiate avalanches of high-energy runaway electrons whose bremsstrahlung, in turn, is capable of accounting for photonuclear reactions. With this scenario, the (γ, n) yield in gigantic stratospheric discharges was calculated analytically to be $N_n \approx 10^{15}$ neutrons per discharge [Babich, 2006, 2007; Babich and Roussel-Dupré, 2007] and obtained from numerical simulations as $N_n \approx 10^{13}$ – 10^{14} neutrons [Babich et al., 2007b, 2007a, 2008a] under the supposition of rather large thundercloud charges. The (γ, n) yield from the channel of intracloud lightning discharge was estimated as $N_n \approx 4 \cdot 10^{13}$ neutrons per discharge [Babich and Roussel-Dupré, 2007].

[4] Absorption of neutrons in dense lower layers of the atmosphere is so strong that the predicted values of neutron yield N_n in stratospheric discharges, even in the field of large thundercloud charges, are insufficient to account for the increases in neutron fluxes observed on the Earth's surface [Shah et al., 1985; Shyam and Kaushik, 1999; Kuzhevskij, 2004; Gusev et al., 2009] by γ emission in the stratosphere [Babich, 2006, 2007; Babich and Roussel-Dupré, 2007]. Assuming a source of γ emission responsible for (γ, n) reactions located in the stratosphere, large delay times of neutron arrival at the detector relative to the lightning EMP t_{del} , detected by Shah et al. [1985], can be explained, but it is impossible to account for $t_{\text{del}} \leq 100 \mu\text{s}$ [Babich, 2006, 2007; Babich and Roussel-Dupré, 2007].

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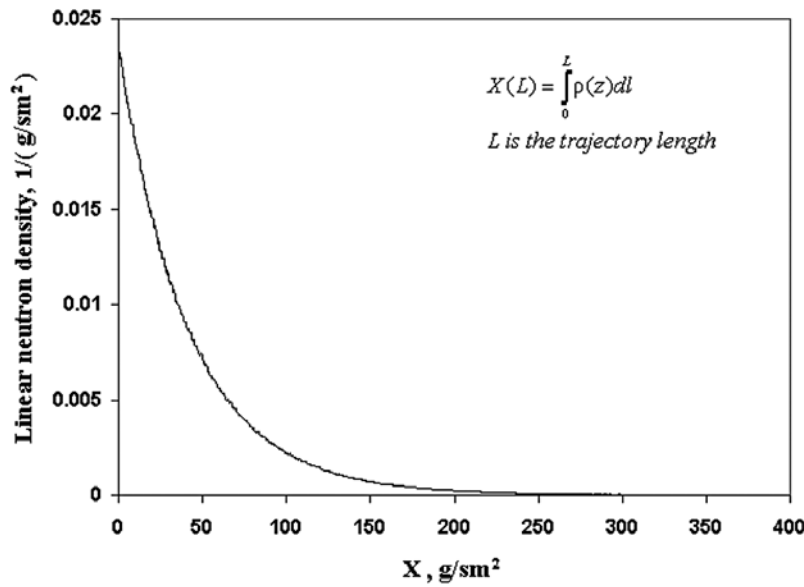


Figure 1. Dependence of the neutron linear density on the quantity of passed air.

Therefore to interpret the thunderstorm correlated neutron increases [Shah *et al.*, 1985; Shyam and Kaushik, 1999; Kuzhevskij, 2004; Gusev *et al.*, 2009] it is necessary to assume a tropospheric source of (γ, n) reactions. This is the case because the hard γ rays were observed to be associated with tropospheric sources [cf., e.g., Eack *et al.*, 1996; Chubenko *et al.*, 2000; Torii *et al.*, 2002; Dwyer *et al.*, 2004; Tsuchiya *et al.*, 2007, 2009, and references therein].

[5] The (γ, n) yield $N_n \approx 4 \cdot 10^{13}$ neutrons estimated for lightning [Babich and Roussel-Dupré, 2007] is sufficient to account for the on-ground thunderstorm correlated neutron increases, especially at the high mountain observatory [Shah *et al.*, 1985]. However, recently Tsuchiya *et al.* communicated an observation of hard γ ray burst at the altitude 2770 m (almost the same altitude as in the neutron observations by Shah *et al.*), the source of which the authors, basing on measurements of electric field and optical emissions, connected with mountain top thunderclouds rather than with lightning [Tsuchiya *et al.*, 2007, 2009]. As the detected γ ray spectrum extends above the photonuclear reaction threshold for nitrogen $\varepsilon_{\gamma,th} = 10.5$ MeV [Dietrich and Berman, 1988], analysis of the tropospheric and volumetric (thundercloud, not lightning) source of photonuclear neutrons seems expedient.

[6] The neutron source localization would allow learning more about the origin of the observed neutron increases. Therefore, to localize the neutron source altitudes, we carried out numerical simulations of volumetric photonuclear source located in the troposphere (thundercloud) and neutron transport from various altitudes to the altitude of the detector location in the mountains [Shah *et al.*, 1985] and to sea level [Shyam and Kaushik, 1999; Kuzhevskij, 2004; Gusev *et al.*, 2009].

2. Photonuclear Neutron Source

[7] The calculation of the photonuclear neutron source was performed for a “cloud altitude” Z_{cl} point γ ray source

emitting photons with energies greater than the (γ, n) threshold for nitrogen $\varepsilon_{\gamma,th} = 10.5$ MeV. Photon transport was simulated by Monte Carlo techniques with allowance for Compton scattering of photons, production of electron-positron pairs, bremsstrahlung of electrons and positrons with cross sections available in [Belen’kii, 1948; Bethe and Ashkin, 1953]. The universal spectrum of bremsstrahlung from relativistic electron avalanches [Babich *et al.*, 2004a] for the γ rays and cross sections of the (γ, n) reactions available in the review [Dietrich and Berman, 1988] were used. As photon energies are so large that the scattering angles are small and secondary electrons and positrons move almost in the directions of primary photon propagation, one-dimensional simulations were conducted, with the number of simulated neutron trajectories being equal to 10^9 . As a result the specific (γ, n) yield was obtained:

$$N_{n1} = 4.3 \times 10^{-3} \text{ neutrons per one photon with energy } \varepsilon_{\gamma} \geq \varepsilon_{\gamma,th}. \quad (1)$$

Figure 1 illustrates a dependence of the linear neutron density normalized per unity on the quantity of air passed by photons $X(L) = \int_0^L \rho(z)dl$ obtained in the result of the simulations. Here L is the length of photon trajectory. This function does not depend on the initial photon altitude and the direction of the primary γ ray flux. Figure 2 displays the energy spectrum of produced neutrons.

3. Simulation of Neutron Transport

[8] Neutron transport in the atmosphere was simulated as follows. We assumed a point source of photons located at some altitude Z_{cl} inside the thundercloud. The source emits uniformly downward into a cone with the angular aperture α . The simulation was carried out using Monte Carlo

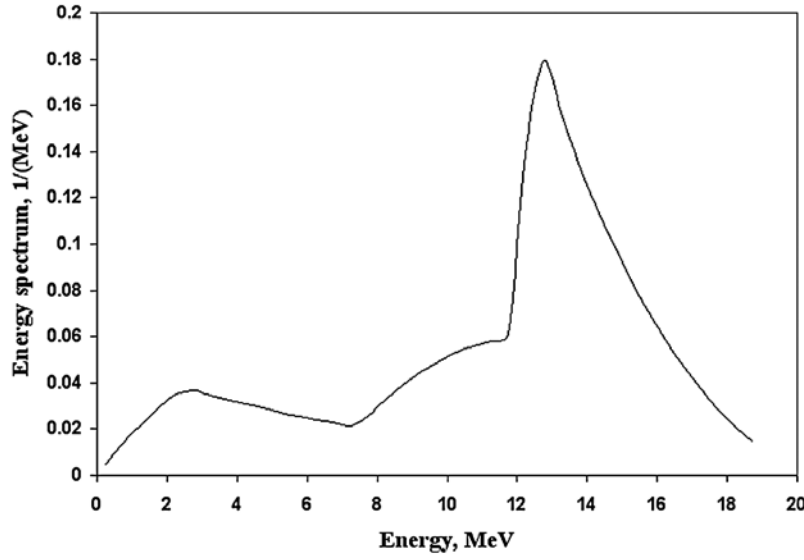


Figure 2. Initial energy distribution of photonuclear neutrons (source).

techniques allowing for elastic scattering (elementary cross section σ_{el}) and neutron capture (elementary cross section σ_c) by a mixture of nitrogen (80%) and oxygen (20%) nuclei. The neutron capture cross section was calculated as the difference between the total cross section and elastic and inelastic scattering cross sections $\sigma_c = \sigma_{tot} - (\sigma_{el} + \sigma_{in})$. The ENDF-7 library (available at <http://www.nndc.bnl.gov/exfor/ENDF00.jsp>) of cross sections was used for these processes.

[9] Let a neutron, initially located at point A, experience an elastic collision at point B and after being scattered at the angle θ appear at point C (see Figure 3). The coordinates of the neutron after the scattering event are as follows:

$$\begin{aligned} x_C &= x_B + BC \cdot (-\sin \theta \cdot \cos \varphi \cdot \cos \phi \cdot \cos \chi + \sin \theta \\ &\quad \cdot \sin \varphi \cdot \cos \psi + \cos \theta \cdot \sin \phi \cdot \cos \chi), \\ y_C &= y_B + BC \cdot (-\sin \theta \cdot \cos \varphi \cdot \cos \phi \cdot \cos \psi - \sin \theta \\ &\quad \cdot \sin \varphi \cdot \cos \chi + \cos \theta \cdot \sin \phi \cdot \cos \psi), \\ z_C &= z_B + BC \cdot (\sin \theta \cdot \cos \varphi \cdot \sin \phi + \cos \theta \cdot \cos \phi), \end{aligned} \quad (2)$$

where

$$\begin{aligned} \cos \chi &= \frac{x_B - x_A}{\sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}}, \\ \cos \psi &= \frac{y_B - y_A}{\sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}}, \\ \cos \phi &= \frac{z_B - z_A}{AB}. \end{aligned}$$

If $|\cos \phi| = 1$ the projection of AB on the plane OXY is equal to zero, and the angles χ and ψ are not defined. In this case instead of (2) from Figure 3 we have

$$\begin{aligned} x_C &= x_B + BC \cdot \sin \theta \cdot \cos \varphi, \\ y_C &= y_B + BC \cdot \sin \theta \cdot \sin \varphi, \\ z_C &= z_B + BC \cdot \cos \theta \cdot \cos \phi. \end{aligned}$$

Here $\varphi \in [0, 2\pi]$ is a random quantity.

[10] The neutron distribution function over the scattering angles is presented as an expansion over the Legendre polynomials P_l :

$$f(\varepsilon, \mu) = \sum_{l=0}^N \frac{2l+1}{2} a_l(\varepsilon) \cdot P_l(\mu), \quad (3)$$

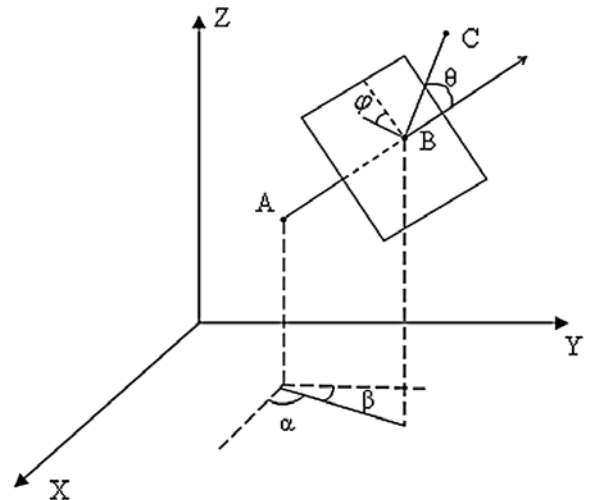


Figure 3. Neutron scattering scheme.

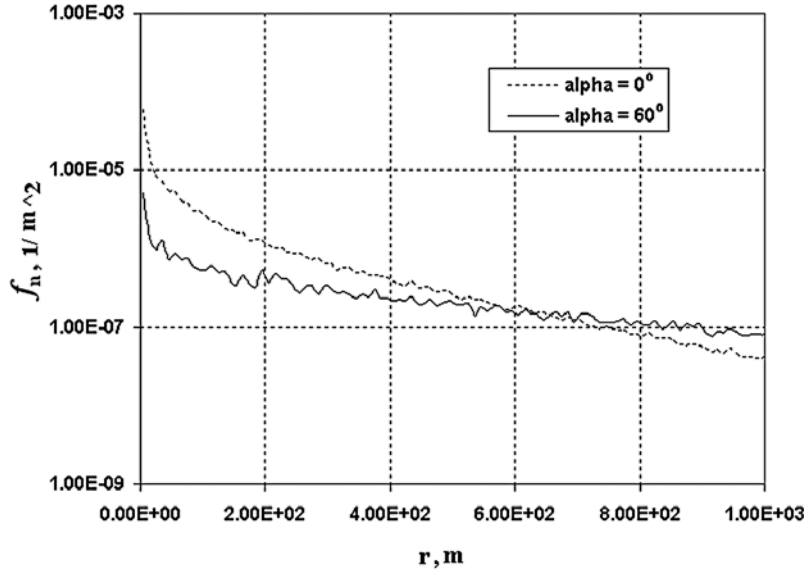


Figure 4. Neutron flux density normalized per unity at the detector altitude ($Z_{\text{det}} = 3$ km) in the case of point γ ray source located at the altitude $Z_{\text{cl}} = 8$ km.

where ε is the neutron energy, μ is the scattering angle cosine in the c.m. reference frame and the coefficients $a_i(\varepsilon)$ are available in the ENDF-7 library. For the neutron scattering angle cosine the average cosine is taken, defined as follows in the laboratory frame of reference:

$$\langle \cos \theta \rangle = \int_{-1}^1 \frac{A\mu + 1}{\sqrt{A^2 + 2A\mu + 1}} f(\mu, \varepsilon) \cdot d\mu, \quad (4)$$

where A is the mass of target nucleus in neutron mass units.

[11] The neutron energy after scattering is connected with the initial neutron energy ε_0 as follows:

$$\varepsilon = \frac{A^2 + 2A\langle \cos \theta \rangle + 1}{(A + 1)^2} \varepsilon_0. \quad (5)$$

The simulation of the neutron trajectory is terminated when the neutron reaches the “detector altitude” $z = Z_{\text{det}}$. The result of the calculation is the neutron flux density $f_n(Z_{\text{cl}}, Z_{\text{det}}, \alpha, r)$ normalized per unity, where r is the distance from the symmetry axis of the problem. In Figure 4 the function $f_n(Z_{\text{cl}}, Z_{\text{det}}, \alpha, r)$ is shown for the case $Z_{\text{det}} = 3$ km, $Z_{\text{cl}} = 8$ km and two values of the angular aperture of the γ flux $\alpha = 0^\circ$ and $\alpha = 60^\circ$.

4. Results and Discussion

[12] We calculated the specific neutron yield and flux density from a point source of γ emission, i.e., bremsstrahlung of runaway electrons (REs). In nature the bremsstrahlung generation is a volumetric process. This fact is to be taken into account in view of the source of the hard γ ray burst with spectrum extending above the (γ, n) threshold, as observed by Tsuchiya *et al.* [2009] on the ground, was most likely distributed inside the thundercloud. For the sake of simplicity we simulated a homogeneous neutron source in the form of a disk with a radius R_{cl} (Figure 5) located at the altitude $z = Z_{\text{cl}}$. The number of

photons emitted by an element $dS' = r'dr'd\varphi'$ of the disk surface is $dN_{\gamma, \text{cl}} = \frac{N_{\gamma, \text{cl}}}{\pi R_{\text{cl}}} r'dr'd\varphi'$, where $N_{\gamma, \text{cl}}$ is the total number of photons emitted by the disk. The number of neutrons produced in the atmosphere due to each elementary photon source capable of reaching the detector surface with the total area S_{det} is calculated as $dN_{n, \text{det}} = \nu(Z_{\text{cl}}, Z_{\text{det}}) \cdot f_n(Z_{\text{cl}}, Z_{\text{det}}, \alpha, r) \cdot S_{\text{det}} \cdot dN_{\gamma, \text{cl}}$, where $\nu(Z_{\text{cl}}, Z_{\text{det}})$ is the neutron number at the altitude Z_{det} per one photon of the source. Integrating this expression over the disk area and dividing the result by $\nu \cdot N_{\gamma, \text{cl}} \cdot S_{\text{det}}$, we obtain the neutron flux density at the detector as a function of the distance r from the disk symmetry axis:

$$F_n(Z_{\text{cl}}, Z_{\text{det}}, \alpha, r) = \frac{1}{\pi R_{\text{cl}}^2} \int_0^{R_{\text{cl}}} \int_0^{2\pi} f_n(Z_{\text{cl}}, Z_{\text{det}}, \alpha, R) \cdot r' d\varphi' dr', \quad (6)$$

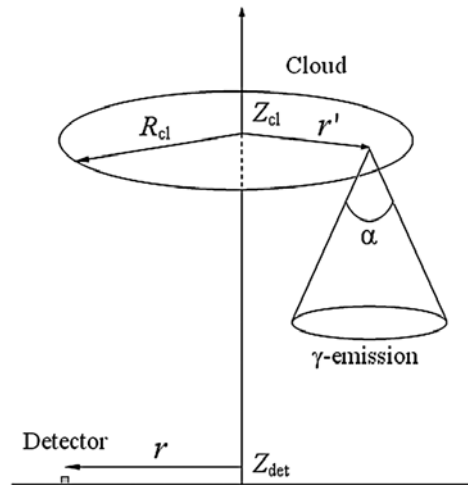


Figure 5. Model geometry for the case of volumetric γ ray source (cloud).

Table 1. Number, Flux Density, and Fluence of Neutrons at the “Detector Altitude” for Neutron Source Located at “Cloud Altitude”

Z_{det} (km)	Z_{cl} (km)	R_{cl} (km)	$\alpha = 0^\circ$			$\alpha = 60^\circ$		
			N_n	F_n^{max} (1/m ²)	$N_n \cdot F_n^{\text{max}}$ (n/m ²)	N_n	F_n^{max} (1/m ²)	$N_n \cdot F_n^{\text{max}}$ (n/m ²)
0	15	1.7	$7.9 \cdot 10^5$	$1.1 \cdot 10^{-7}$	0.09	$3.1 \cdot 10^5$	$0.86 \cdot 10^{-7}$	0.03
0	12	1.5	$2.4 \cdot 10^6$	$1.4 \cdot 10^{-7}$	0.34	$1.5 \cdot 10^6$	$0.57 \cdot 10^{-7}$	0.09
0	10	1.2	$1.1 \cdot 10^7$	$2.1 \cdot 10^{-7}$	2.30	$6.3 \cdot 10^6$	$0.73 \cdot 10^{-7}$	0.46
0	8	1.0	$5.5 \cdot 10^7$	$2.9 \cdot 10^{-7}$	16.00	$3.5 \cdot 10^7$	$1.3 \cdot 10^{-7}$	4.55
0	5	1.0	$2.4 \cdot 10^9$	$2.9 \cdot 10^{-7}$	696	$1.0 \cdot 10^9$	$1.7 \cdot 10^{-7}$	170
3	12	1.5	$1.2 \cdot 10^9$	$1.3 \cdot 10^{-7}$	156	$5.5 \cdot 10^8$	$0.64 \cdot 10^{-7}$	35.2
3	10	1.2	$3.0 \cdot 10^9$	$1.9 \cdot 10^{-7}$	570	$3.4 \cdot 10^9$	$0.96 \cdot 10^{-7}$	326.4
3	8	1.0	$1.5 \cdot 10^{10}$	$2.6 \cdot 10^{-7}$	3900	$1.85 \cdot 10^{10}$	$1.5 \cdot 10^{-7}$	2775

where

$$R = \sqrt{r^2 + r'^2 - 2rr' \cdot \cos \varphi'}.$$

To calculate the absolute number of generated neutrons, the number of photons $N_\gamma(\varepsilon_\gamma \geq \varepsilon_{\gamma,\text{th}})$ with energies above the (γ, n) threshold $\varepsilon_{\gamma,\text{th}} = 10.5$ MeV is required. To estimate the $N_\gamma(\varepsilon_\gamma \geq \varepsilon_{\gamma,\text{th}})$, we used the integral characteristic of Res

$$g = \int_{H_{\text{cl}}}^{H_{\text{max}}} g(z) dz = 2\pi \int_{H_{\text{cl}}}^{H_{\text{max}}} dz \int_0^{t_{\text{max}}} dt \int_0^{r_{\text{max}}} n_{\text{run}}(z, r, t) r dr \approx 10^{11}, \quad (7)$$

where the values $H_{\text{max}} = 74$ km, $r_{\text{max}} = 30$ km and $t_{\text{max}} = 3.5$ ms constrain the calculation area in the vertical z and horizontal r directions and the duration of the simulations of the discharge driven by REs [Babich et al., 2008a], and reduced to 1 atm. bremsstrahlung generation rate per one RE [Babich et al., 2004a]:

$$\frac{1}{N_e(t)} \left(\frac{dN_\gamma(t)}{dt} \right)_{\text{emis}} \approx 0.01 / (\text{ns atm.}). \quad (8)$$

The portion of photons with energies above $\varepsilon_{\gamma,\text{th}}$ received from the universal RE bremsstrahlung spectrum, calculated in [Babich et al., 2004a], is $\Delta \approx 0.008$. Then at the pressure P we obtain

$$N_\gamma(\varepsilon_\gamma \geq \varepsilon_{\gamma,\text{th}}) \approx \frac{gP\Delta}{N_e(t)} \left(\frac{dN_\gamma(t)}{dt} \right)_{\text{emis}} \approx 10^{16} \cdot P(z) \text{ photons}. \quad (9)$$

Assuming here $P \geq 0.1$ atm., i.e., altitudes $z \leq 16$ km, we obtain $N_\gamma(\varepsilon_\gamma \geq \varepsilon_{\gamma,\text{th}}) \geq 10^{15}$ and, consequently, allowing for (1) we obtain the initial number of neutrons in the source distributed over the altitudes (quantity of the air passed by photons) as in Figure 1:

$$N_n(0) = N_{n1} \cdot N_\gamma(\varepsilon_\gamma \geq \varepsilon_{\gamma,\text{th}}) \geq 4.3 \cdot 10^{12}. \quad (10)$$

[13] Presented in Table 1 are the calculated neutron numbers N_n , peak value of the flux density $F_n(Z_{\text{cl}}, Z_{\text{det}}, \alpha, r = 0) = F_n^{\text{max}}$, and the corresponding neutron fluence at the “detector altitude” Z_{det} for $N_n(0) = 4.3 \cdot 10^{12}$ according to (10), two values of the γ flux angular aperture α and various $Z_{\text{cl}}, Z_{\text{det}}$ and R_{cl} . Figure 6 shows the neutron flux density for

the case $Z_{\text{cl}} = 8$ km, $Z_{\text{det}} = 3$ km, $R_{\text{cl}} = 1$ km, $\alpha = 0^\circ$ and $\alpha = 60^\circ$. With the aperture $\alpha = 0^\circ$ the flux is more concentrated close to the symmetry axis. Figure 7 shows the energy distribution of all neutrons for $Z_{\text{cl}} = 8$ km, $Z_{\text{det}} = 3$ km, $R_{\text{cl}} = 1$ km and $\alpha = 60^\circ$.

[14] Shah et al. [1985] ($Z_{\text{det}} \approx 3$ km) detected events with the number of neutrons arriving at the detector $N_{\text{det}} = 3\text{--}60$. With the detector efficiency $\eta \sim 3\%$ and area $S_{\text{det}} \approx 3$ m² these numbers N_{det} correspond to the neutron fluence N_{det}/η $S_{\text{det}} \approx (30\text{--}670)$ n/m², which is in agreement with the calculated fluence $N_n \cdot F_n^{\text{max}}$ at $r = 0$ (peak) for the cloud altitudes Z_{cl} ranging from 8 to 12 km.

[15] Unfortunately comparison of the results of our simulations with the data of the experiments executed close to the sea level [Shyam and Kaushik, 1999; Kuzhevskij, 2004; Gusev et al., 2009] is impossible due to the lack of required details in these papers. The this same reason we did not simulate the neutron transport to a space based detector for comparison with the observations reported by Bratolyubova-Tsulukidze et al. [2004]. However, we note that for the case $Z_{\text{det}} \approx 0$ km and $Z_{\text{cl}} = 5$ km the calculated fluence is big enough to testify to the fact itself of the neutron flux increases at the sea level. However, it is not clear if a sufficiently intensive γ ray source (RE bremsstrahlung) with photon energies $\varepsilon_\gamma \geq \varepsilon_{\gamma,\text{th}}$ is possible to be produced in the vast domain of thunderstorm atmosphere at such low altitudes.

[16] Additionally we conducted calculations for the case of an atmosphere containing 5% hydrogen (water vapor). Inclusion of hydrogen does not substantially affect the results. Hence, variations of water vapor in the atmosphere do not affect photoneutron generation.

5. Conclusions

[17] To interpret the results of the on-ground observations of neutron flux increases correlated with thunderstorm activity, the generation of photonuclear neutrons and subsequent transport in the atmosphere were simulated. For the primary γ ray source the universal spectrum of relativistic runaway electron avalanche bremsstrahlung [Babich et al., 2004a] was used. The calculations were conducted for a downward directed γ ray flux as assumed by Tsuchiya et al. [2009] while interpreting their observation of a hard γ ray burst. This suggests implicitly the development of descending avalanches of relativistic electrons. It is pertinent to note that the characteristics of the terrestrial γ ray flashes detected in the near space aboard CGRO [Fishman et al.,

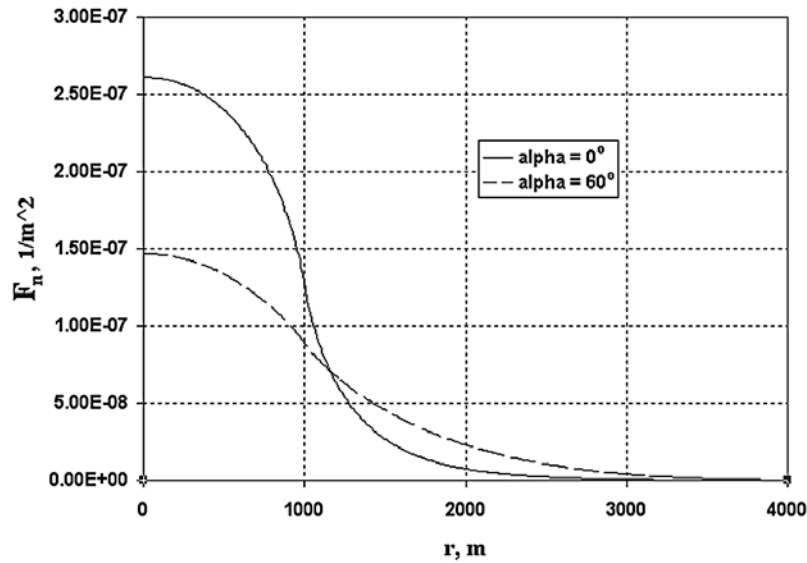


Figure 6. Neutron flux density normalized per unity at the detector altitude ($Z_{\text{det}} = 3$ km) in the case of disk with radius $R_{\text{cl}} = 1$ km located at the altitude $Z_{\text{cl}} = 8$ km modeling the volumetric γ ray source.

1994] and RHESSI [Smith *et al.*, 2005] are explained more or less adequately by bremsstrahlung of ascending relativistic electron avalanches (for instance, [Dwyer and Smith, 2005; Dwyer, 2008; Cummer *et al.*, 2005; Williams *et al.*, 2006; Babich *et al.*, 2007b; 2008a, 2008b, 2008c]), as in the earlier works, where the CGRO data were analyzed [e.g., Taranenko and Roussel-Dupre, 1996; Roussel-Dupre *et al.*, 1998; Lehtinen *et al.*, 1996, 1999, 2001; Gurevich and Zybin, 2001; Babich *et al.*, 2001, 2004b].

[18] The neutron fluence dependence at the altitude 3 km (close to the detector altitude in the work of Shah *et al.* [1985]) on the altitude of the primary photon source Z_{cl} ,

its size and the angular aperture of the γ ray flux were analyzed. The fluence calculated for the altitudes $Z_{\text{cl}} = 8$ –12 km is consistent with the measured neutrons numbers [Shah *et al.*, 1985]. This altitude range is consistent with a volumetric intracloud source of hard γ rays detected by Tsuchiya *et al.* [2009].

[19] In view of the fact that the γ ray burst observed by Tsuchiya *et al.* [2009], was not correlated with lightning, with a high degree of confidence the photonuclear reactions caused by bremsstrahlung of descending relativistic runaway electron avalanches inside thunderclouds rather than in lightning account for the neutron flux increases observed

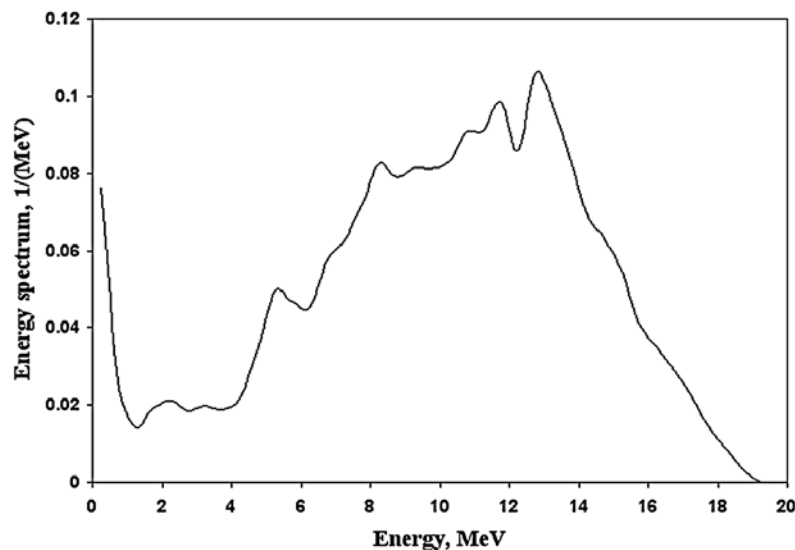


Figure 7. Neutron energy distribution at the detector altitude ($Z_{\text{det}} = 3$ km) in the case of disk with radius $R_{\text{cl}} = 1$ km ($\alpha = 60^\circ$) located at the altitude $Z_{\text{cl}} = 8$ km modeling the volumetric γ ray source.

on the ground in the thunderstorm atmosphere [Shah et al., 1985; Shyam and Kaushik, 1999; Kuzhevskij, 2004; Gusev et al., 2009].

[20] **Acknowledgments.** Amitava Bhattacharjee thanks Lev Dorman and another reviewer for their assistance in evaluating this paper.

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